

Sri Lanka Institute of Information Technology



IT3012 – Intelligent Agents
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Practical Answer Submission

Lab Practical Sheet No: 01

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Part 1

Task 1.1

1.
 - P - Performance Measure
 - E - Environment
 - A - Actuators
 - S - Sensors
2.
 - self.width
 - self.height
 - self.walls
 - self.food_positions
 - self.opponents
 - self.agent_pos
3. Multi-Agent environment

This is a Multi-Agent environment because there are multiple agents interacting in the same environment (the player agent and the opponents).

Task 1.2

4. A percept sequence is the complete history of all percepts (sensor information) received by an agent over time through its sensors.
5. Sensors (S)
6. The agent receives only agent position, opponent position, food smell, wall detection, collision and score. But it doesn't receive complete map, wall locations, food locations or future opponent movements. It only knows what the sensor tells it. Therefore, Partially Observable.

Task 1.3

7. Performance Measure (P)
8. It is structurally important that the performance metric evaluates changes in the external environment because intelligent agents are measured by the results of their actions rather than their internal processing. Hitting a wall is a negative environmental outcome that prevents effective movement and progress toward goals. By penalizing such actions, the performance measure accurately reflects the agent's success in interacting with the environment. This supports the concept of rational agents, which aim to maximize their performance through effective actions and outcomes.

Part 2

Step 2.1

9. The environment becomes Partially Observable because the toxic traps exist in the environment, but the agent's sensors do not provide any information about them. The agent cannot perceive all aspects of the environment, so it has incomplete knowledge when making decisions.

Step 2.2

10. The new smells_toxin sensor provides additional information about dangerous locations. This expands the agent's percept sequence, allowing it to avoid toxic traps and choose safer actions. As a result, the agent can maximize its expected utility (score) by making more rational decisions based on better information.

Step 2.3

11. The classic Vacuum World exploit was an agent that repeatedly cleaned the same dirt (or repeatedly created and cleaned dirt) to gain points without actually keeping the environment clean. This demonstrated that a poorly designed performance measure can be exploited, so performance measures should reward achieving the intended goal rather than allowing repetitive actions to accumulate rewards.